

23rd Experiment

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import accuracy_score
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Generate Spiral Dataset
def generate_spiral_data(points=500):
    np.random.seed(42)

    n = points // 2

    theta = np.sqrt(np.random.rand(n)) * 2 * np.pi
    r = 2 * theta + np.pi

    x1 = np.array([np.cos(theta) * r,
                   np.sin(theta) * r]).T

    x2 = np.array([np.cos(theta + np.pi) * r,
                   np.sin(theta + np.pi) * r]).T

    X = np.vstack([x1, x2])
    y = np.hstack([np.zeros(n), np.ones(n)])

    return X, y

X, y = generate_spiral_data(500)
```

```
# Build Neural Network

model = Sequential([
    Dense(16, activation='sigmoid', input_shape=(2,)),
    Dense(8, activation='sigmoid'),
    Dense(1, activation='sigmoid')
])
```

```
# Compile Model

model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

```
# Train Model

history = model.fit(
    X, y,
    epochs=100,
    verbose=0
)
```

```
# Training Loss Graph

plt.figure(figsize=(5,3))
plt.plot(history.history['loss'])
plt.title("Training Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.grid(True)
plt.show()
```

```
# Prediction

pred = model.predict(X)

pred_class = (pred > 0.5).astype(int)

# Accuracy

acc = accuracy_score(y, pred_class)

print("Accuracy:", round(acc*100,2), "%")
```

```
# Plot Spiral Dataset

plt.figure(figsize=(5,4))

plt.scatter(X[:,0], X[:,1], c=y, s=15)

plt.title("Spiral Dataset")

plt.xlabel("Feature 1")

plt.ylabel("Feature 2")

plt.show()
```

